

1.

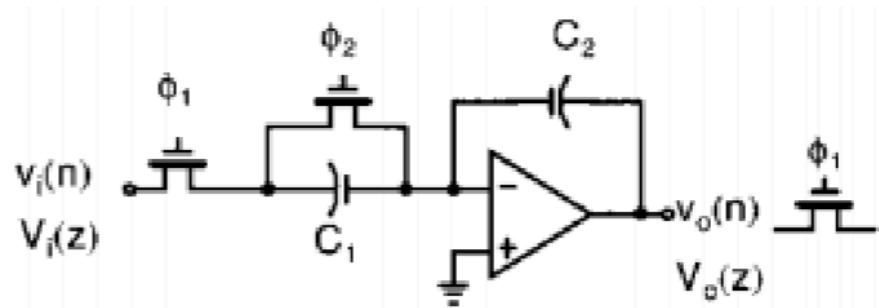


Fig. 1

(a) Neglecting parasitic capacitances, find the discrete time transfer function of the circuit in Fig. 1. Assume that the op amp is ideal

(10 marks)

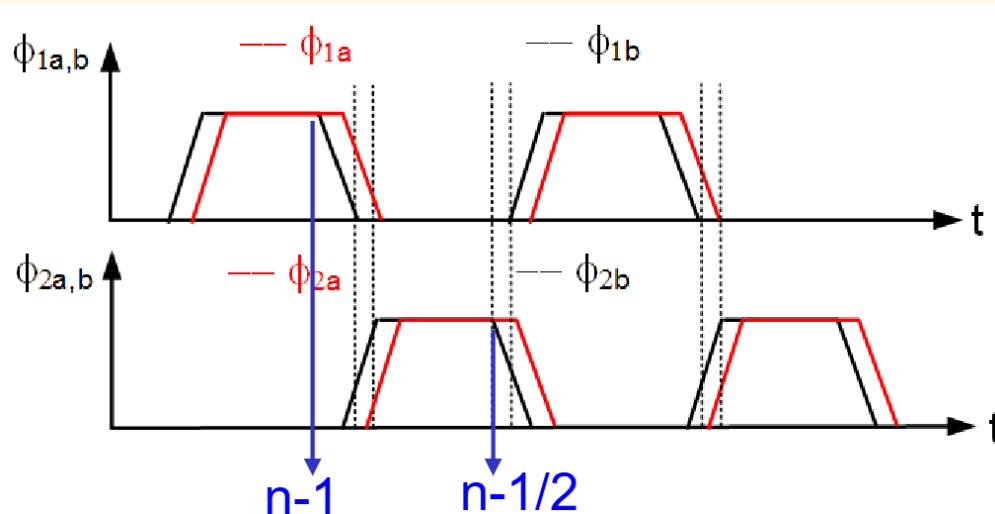
(b) Including parasitic capacitances, find the discrete time transfer function of the circuit in Fig. 1

(10 marks)

1. Is the floating MOS (red boxed) supposed to be connected to anything?**

ANSWER: It is a convention to indicate that the signal is only sampled (by next stage) during ϕ_1

2. May we have the ϕ_1 ϕ_2 timing diagram, please?** ANSWER: These are the standard non-overlapping $\phi_1, 2$ clocking signals. You can refer to Lecture 11 slide 26 for the timing diagram.



2. Consider delay cell based ring oscillator shown in [Fig. 2a](#). With delay stages shown in [Fig. 2b](#)

Delay cell based Oscillator

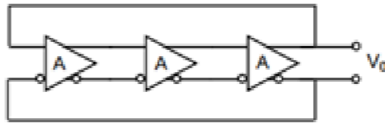


Fig. 2a

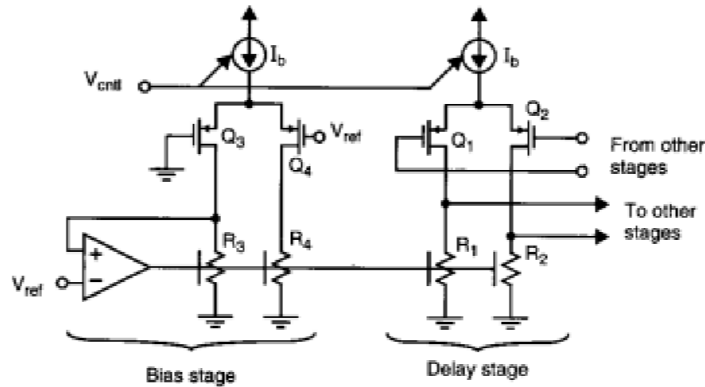


Fig. 2b

Derive the oscillation frequency equation with the following assumptions

- (1) Capacitive load of each delay stage is CL
- (2) When the delay cell switches, the bias current I_b changes quickly from one side to the other.
- (3) Small differential voltage can switch the delay cell

(20 marks)